

RaceDay REST API Endpoint Plan

PROG6212 POE — Part 1: System Planning and Database

RaceDay Event Management System

Sibusiso Nahara – ST10482952

Introduction

This endpoint plan defines the REST API that will support the RaceDay event management system. The API will provide authentication, user profile management, event management, category management, event enrolments and result management. Role-based access will be applied to protect organiser and participant functionality. The endpoint plan will be used as the specification for the API developed in Part 2.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new user account and assigns the selected Organiser or Participant role.	Public	{ "firstName": "John", "lastName": "Smith", "email": "john@example.com", "password": "Password123!", "role": "Participant", "phoneNumber": "0821234567" }	201 Created { "message": "Registration successful.", "userId": 5 }
POST	/api/auth/login	Authenticates a registered user and establishes the authenticated session and role.	Public	{ "email": "john@example.com", "password": "Password123!" }	200 OK { "message": >Login successful.", "userId": 5, "role": "Participant" }
GET	GET /api/users/me	Returns the authenticated	Organiser /	None	200 OK { "userId": 5,

		authenticated user's profile information.	Participant		<pre> "firstName": "John", "lastName": "Smith", "email": "john@example.com", "role": "Participant", "phoneNumber": "0821234567" </pre>
PUT	PUT /api/users/me	Updates the authenticated user's personal profile information.	Organiser / Participant	<pre> { "firstName": "John", "lastName": "Smith", "phoneNumber": "0839876543" } </pre>	200 OK <pre> { "message": "Profile updated successfully." } </pre>
GET	GET /api/events	Returns upcoming RaceDay events that participants can browse.	Participant / Organiser	None	200 OK <pre> [{ "eventId": 1, "eventName": "Durban Road Challenge", "eventDate": "2026-10-10", "startTime": "07:00", "location": "Durban", "status": "Upcoming" }] </pre>
GET	GET /api/events/{eventId}	Returns detailed information about a specific event, including its availability.	Participant / Organiser	None	200 OK <pre> { "eventId": 1, "eventName": "Durban Road Challenge", "description": "Annual road </pre>

		e categories.			running event.", "eventDate": "2026-10-10", "startTime": "07:00", "location": "Durban", "categories": [] }
POST	POST /api/events	Creates a new RaceDay event.	Organiser	{ "eventName": "Durban Road Challenge", "description": "Annual road running event.", "eventDate": "2026-10-10", "startTime": "07:00", "location": "Durban", "entryFee": 150.00 }	201 Created { "eventId": 1, "message": "Event created successfully." }
PUT	PUT /api/events/{eventId}	Updates an event managed by the authenticated organiser.	Organiser	{ "eventName": "Durban Road Challenge Updated", "description": "Updated event information.", "eventDate": "2026-10-17", "startTime": "07:30", "location": "Durban", "entryFee": 175.00 }	200 OK { "message": "Event updated successfully." }
DELETE	DELETE /api/events/{eventId}	Deletes an event managed by the authenticated	Organiser	None	204 No Content

		organiser.			
POST	POST /api/events/{eventId}/categories	Creates an age or distance category for an event.	Organiser	{ "categoryName": "10 km", "distanceKm": 10, "description": "10 kilometre road race.", "entryFee": 150.00, "maxParticipants": 500 }	201 Created { "categoryId": 1, "message": "Category created successfully." }
GET	GET /api/events/{eventId}/categories	Returns all categories available for a specific event.	Organiser / Participant	None	200 OK [{ "categoryId": 1, "categoryName": "10 km", "distanceKm": 10, "availableSlots": 250 }]
PUT	PUT /api/events/{eventId}/categories/{categoryId}	Updates a category belonging to an organiser's event.	Organiser	{ "categoryName": "10 km Road Race", "distanceKm": 10, "description": "Updated category.", "entryFee": 175.00, "maxParticipants": 600 }	200 OK { "message": "Category updated successfully." }
DELETE	DELETE	Deletes a	Organiser	None	204 No Content

	/api/events/{eventId}/categories/{categoryId}	category from an organiser's event when it is not being used by enrolments.			
POST	POST /api/enrolments	Enrols the authenticated participant in a selected event category.	Participant	{ "eventCategoryId": 3 }	201 Created { "enrolmentId": 12, "message": "Enrolment created successfully." }
GET	GET /api/enrolments/me	Returns all event enrolments belonging to the authenticated participant.	Participant	None	200 OK [{ "enrolmentId": 12, "eventName": "Durban Road Challenge", "categoryName": "10 km", "enrolmentDate": "2026-09-01", "raceNumber": "R1025", "status": "Active" }]
GET	GET /api/events/{eventId}/enrolments	Returns all participant enrolments for an event managed by the	Organiser	None	200 OK [{ "enrolmentId": 12, "participantName": "John Smith",

		organiser.			"categoryName": "10 km", "raceNumber": "R1025", "status": "Active" }]
POST	POST /api/enrolments/{enrolmentId}/result	Records a participant's finishing time and finishing position.	Organiser	{ "finishTime": "01:02:35", "position": 14, "resultStatus": "Finished" }	201 Created { "resultId": 25, "message": "Result recorded successfully." }
GET	GET /api/results/me	Returns the authenticated participant's personal race results.	Participant	None	200 OK [{ "eventName": "Durban Road Challenge", "categoryName": "10 km", "finishTime": "01:02:35", "position": 14, "resultStatus": "Finished" }]
GET	GET /api/events/{eventId}/results	Returns recorded results for an event managed by the organiser.	Organiser	None	200 OK [{ "participantName": "John Smith", "categoryName": "10 km", "finishTime": "01:02:35", "position": 14

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